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Lean Manufacturing for Improved Overall Equipment Effectiveness and Reduced Carbon Dioxide Emissions in Beverage Production

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Abstract: The energy intensity of beverage manufacturing creates substantial opportunities for simultaneous improvements in operational efficiency and environmental performance. In this study, the association between overall equipment effectiveness (OEE) and energy consumption and carbon dioxide (CO_2) emissions intensity was evaluated in a beverage manufacturing facility in Bangladesh following the implementation of lean manufacturing practices. Real-time production data were analyzed to identify major sources of equipment inefficiency, after which total productive maintenance, Sort, Set in Order, Shine, Standardize, and Sustain (5S), and single-minute exchange of die were implemented to address equipment downtime, operational inefficiencies, quality losses, and changeover-related disruptions. Before implementation, an average OEE of 45.15% and a CO_2 emissions intensity of 15.14 g CO_2 per product were recorded. Following the lean interventions, the average OEE increased from 45.15% to 54.12%, reflecting a 19.8% improvement. Additionally, CO_2 emissions intensity decreased from 15.14 to 13.65 g per product, corresponding to a 9.8% reduction. Improvements were observed across availability, performance, and quality, with the reduction in equipment downtime associated with total productive maintenance coinciding with the increase in OEE. Reduced downtime, fewer quality-related losses, and more stable production were associated with lower energy consumption per unit of acceptable product and lower specific CO_2 emissions. The findings indicate that operational efficiency and environmental performance improved concurrently following the implementation of OEE-oriented lean manufacturing practices. Particular attention to equipment availability, material quality, changeover efficiency, and operator training was associated with simultaneous improvements in productivity and carbon intensity. The proposed approach provides a practical framework for integrating lean manufacturing with energy-efficiency and decarbonization objectives in beverage manufacturing and other energy-intensive production environments.

Keywords: Overall equipment effectiveness; Lean manufacturing; Carbon dioxide emissions; Energy efficiency; Sustainable production; Beverage industry

1 Introduction

The global beverage industry, valued at over $1.5 trillion, encompasses a range of products, including bottled water and energy drinks. Its rapid expansion has led to increased production demands and significant environmental impacts, particularly greenhouse gas emissions, including carbon dioxide (CO_2). The industry's energy-intensive operations—such as heating, cooling, and bottling—rely heavily on fossil fuels, underscoring its contribution to climate change. In 2022, the industrial sector was accountable for nearly 38% of global fuel combustion emissions, with a remarkable 70% increase in emissions since 2000, driven by growing consumer demand [1]. In Bangladesh, the food and beverage industry has expanded significantly in recent decades. It accounts for 22% of Bangladesh's manufacturing sector [2]. Urbanization has significantly increased demand in the beverage sector, driven by rising income levels. Within manufacturing, electricity is essential for operating equipment, while thermal energy is utilized for cooking processes. Notably, breweries account for 70% of their electricity usage on refrigeration, packaging, and compressed air tools. Emphasizing energy conservation not only helps to reduce costs and minimize waste but also

fosters a healthier environment by prioritizing efficiency [3]. The energy consumed by refrigeration contributes to greenhouse gas emissions, primarily when powered by fossil fuels, highlighting the need for decarbonization [4].

The beverage industry contributes significantly to CO_2 emissions because of its energy-intensive operations. In 2022, Bangladesh's electricity and heat production sector accounted for approximately 58% of the country's energy-related CO_2 emissions [5]. Consequently, improving energy efficiency in beverage manufacturing is increasingly important for both industrial competitiveness and environmental sustainability. Beverage bottling lines in Bangladesh face significant OEE constraints, more severe than those in industrialized countries. According to the Bangladesh Power Development Board in 2023, frequent voltage fluctuations and unplanned load-shedding events disrupt packaging processes and can lead to substantial production downtime. Additionally, inconsistent preventive and predictive maintenance due to limited access to spare parts and training results in reactive problem-solving after breakdowns. These issues contribute to a baseline OEE that is far below the world-class benchmark of 85%. OEE is crucial for enhancing manufacturing efficiency in the beverage industry, supporting competitiveness and sustainability. The Bangladesh Power Development Board also reported in 2023 that Bangladesh's electricity generation is mainly dependent on natural gas, with furnace oil and diesel accounting for more than 80% of power production. In addition, the Sustainable and Renewable Energy Development Authority noted in 2022 that the country faces energy shortages due to growing industrial demand and limited resources, highlighting the importance of improving energy efficiency in manufacturing. Improving OEE in the beverage sector is essential for enhancing operational performance, minimizing downtime, and maximizing resource use [2]. In a gas-dependent energy system, this enhancement helps reduce energy waste, whereas increased efficiency may reduce the pressure on the power grid, decrease electricity consumption and lower greenhouse gas emissions [6]. The implementation of optimizing processes enables companies to sustain production levels while simultaneously decreasing energy consumption. The adoption of energy-efficient practices within the beverage industry not only contributes to a reduction in carbon dioxide emissions but also aligns with Bangladesh's commitments under the Paris Agreement. This dual benefit underscores the importance of integrating sustainability measures in industrial operations to enhance environmental stewardship and comply with international climate agreements. The country aims to reduce overall emissions by 5% unconditionally by 2030. As climate change grows into a global crisis, industries must adopt sustainable practices and reduce their carbon footprints, following agreements like the Paris Accord. A key focus of this effort is enhancing OEE, which optimizes productivity and reduces energy use, thereby minimizing emissions, making it vital for sustainable growth in the beverage sector.

Although sustainability and operational efficiency have both received considerable research attention, the existing literature reveals a significant conceptual separation between these research domains. Most OEE-related studies focus on productivity enhancement, downtime reduction, maintenance optimization, and lean implementation, whereas environmental studies typically examine energy consumption, carbon accounting, and sustainability indicators independently. As a result, the operational mechanisms through which improvements in OEE influence CO_2 emissions remain insufficiently understood, particularly in beverage manufacturing, where refrigeration, compressed air systems, and packaging operations create substantial electricity demand. A limited number of studies have explored the relationship between equipment effectiveness and environmental performance; however, these studies generally rely on conceptual indicators such as Overall Environmental Equipment Effectiveness (OEEE), are conducted in industries other than beverage manufacturing, or do not incorporate real production-line electricity and emission data. Consequently, there is still a lack of empirical evidence demonstrating how lean-driven improvements in equipment availability, performance, and quality translate into measurable CO_2 emission reductions in an actual beverage production environment.

This study addresses that gap by proposing and evaluating an integrated operational–environmental framework that combines OEE assessment, Pareto-based downtime analysis, Fishbone root-cause identification, and lean manufacturing interventions (Total Productive Maintenance (TPM), Sort, Set in Order, Shine, Standardize, and Sustain (5S), and Single-Minute Exchange of Die (SMED)) with electricity-based CO_2 emission quantification. Using operational data from a beverage packaging line equipped with KHS machinery in Bangladesh, the study examines the association between targeted lean improvements, changes in OEE, and per-unit CO_2 emissions. The originality of this research lies in integrating operational efficiency analysis, lean implementation, and quantitative carbon emission assessment within a single empirical framework, thereby providing practical evidence on the concurrent changes in manufacturing efficiency and environmental performance in high-energy beverage production systems.

2 Literature Review

Previous research related to overall equipment effectiveness (OEE), lean manufacturing, and CO_2 emission reduction can be grouped into two broad research streams. The first stream focuses on operational efficiency, including OEE measurement, total productive maintenance, single-minute exchange of die, 5S, statistical process control, and data-driven productivity improvement. The second stream focuses on environmental sustainability, including energy

management, carbon emission assessment, and green manufacturing indicators. Table 1 summarizes representative studies from both streams and highlights the limited integration between operational performance and environmental assessment in the existing literature.

Table 1. Summary of existing studies on OEE and CO_2 emissions

Method/Tool	OEE/ Efficiency	CO_2 Emission	Reference
Response surface methodology and central composite design	Yes	No	[7]
Pareto/fishbone tools	Yes	No	[8]
Lean	Yes	No	[9]
Lean/total productive maintenance	Yes	No	[10]
Productivity improvements in packaging	Yes	No	[11]
Small food industry sustainability challenges	No	Yes	[12]
Total productive maintenance	Yes	No	[13]
Single-minute exchange of die	Yes	No	[14]
OEE	Indirectly (Paper Industry)	Yes	[15]
Overall environmental equipment effectiveness	No	Yes	[16]
Data analysis in bottling	Yes	No	[17]
Simulation & bottling line optimization	Yes	No	[18]
Industry 4.0 in small and medium-sized enterprises	Yes	No	[19]
Data analytics in brewing	Yes	No	[20]
Single-minute exchange of die	Yes	No	[21]
Statistical process control	No	Yes	[22]
Raw material improvements for sustainability	Yes	No	[23]
Pareto/fishbone/automation tools	No	Yes	[24]
Maynard operation sequence technique	No	Yes	[25]
Brewing industry energy reduction	Yes	No	[26]
Total productive maintenance and statistical process control integration	Yes	No	[27]
Downtime analysis in the beverage industry	No	Yes	[28]
Energy loss categorization	Yes	No	[29]

Note: OEE = overall equipment effectiveness; CO_2 = carbon dioxide.

The studies summarized in Table 1 reveal a clear distinction between research focused on equipment effectiveness and research focused on carbon emission assessment. Most OEE and lean manufacturing studies investigate downtime reduction, maintenance improvement, process optimization, and productivity enhancement, using indicators such as availability, performance, quality, or overall equipment utilization [11, 14]. In contrast, sustainability-oriented studies primarily evaluate energy consumption, greenhouse gas emissions, environmental efficiency, or low-carbon manufacturing strategies without incorporating detailed OEE analysis [12, 22, 25]. A few studies have attempted to connect operational performance with environmental performance. Cercós et al. [15] introduced the concept of OEEE, while Calvo and Domingo [24] discussed the use of CO_2 emissions as a maintenance-related indicator. However, these studies were conducted outside the beverage manufacturing context and did not develop an integrated framework combining root-cause analysis, lean implementation, OEE improvement, and quantitative CO_2 emission measurement based on real industrial operating data.

Consequently, the literature still lacks a comprehensive methodology that simultaneously: (i) identifies the major sources of equipment losses, (ii) applies targeted lean interventions to improve OEE, and (iii) quantifies changes in CO_2 emissions observed at the production-line level. This gap is particularly important for beverage manufacturing, where energy-intensive packaging operations make downtime and process inefficiencies directly relevant to carbon emissions. The present study addresses this gap by integrating OEE evaluation, Pareto analysis, fishbone root-cause identification, total productive maintenance, 5S, single-minute exchange of die, and electricity-based CO_2 emission assessment within a single empirical framework validated on a commercial beverage production line. OEE and total productive maintenance are crucial for enhancing equipment performance and operational efficiency. In recent years, OEE has established itself as a key tool in logistics and services for enhancing performance, measuring productivity, minimizing waste, and improving overall business effectiveness [30]. OEE is key to improving resource efficiency in complex production systems. It enhances bottling line management by providing insights that support decision-making and reduce production failures through total productive maintenance [31].

Statistical process control reduces defects, and total productive maintenance enhances machine availability, jointly improving OEE by over 6% [21, 26]. These examples demonstrate how OEE and total productive maintenance reveal hidden production capacity and enhance reliability across various sectors [32]. Strategic maintenance and adjustments are vital for OEE, primarily influenced by availability, along with performance and quality. Equipment availability can be improved by optimizing maintenance, upgrading machinery, and addressing quality issues [8]. A plan for improvement includes employee training, adherence to standard operating procedures, and regular equipment monitoring [11]. Furthermore, the incorporation of employee training and structured frameworks has effectively closed the gaps in measurement and implementation [13].

The OEE for coffee and goat milk powder sachet packaging was below the 85% benchmark due to equipment failures and inefficiencies [11]. The desirability function and OEE were analyzed using a mathematical model to optimize fruit juice production and enhance production efficiency and product quality [33]. Downtime from the inkjet printer and filler in beverage manufacturing impacts productivity, highlighting the need to assess polyethylene terephthalate bottle filling inefficiencies [27]. In the automotive industry, die preparation times improved, increasing equipment availability from 63.3% to 67.8% and OEE from 60.7% to 65.3% through total productive maintenance techniques [34]. Single-minute exchange of die has improved productivity by 25.63% and increased OEE to 68.41% in industries like beverages and injection molding [14, 20].

Recent research on lean and total productive maintenance indicated improved productivity and reduced waste, resulting in higher yields [10]. Lean practices like value stream mapping and the 5S method have reduced waste and enhanced efficiency, especially in energy-intensive sectors such as food processing and bottling [29, 35]. These integrated practices connect environmental goals with manufacturing efficiency, creating pathways for long-term industrial sustainability [36, 37]. Integrating lean manufacturing with machine learning can enhance production by minimizing changeover times and reducing waste, boosting efficiency and productivity [9]. Growing demand requires modern information technology in packaging. Lean techniques like single-minute exchange of die reduce changeover times, while Tecnomatix software helps identify issues and improves productivity in brewing and packaging [14, 17]. Data-driven approaches and automation enhance equipment management and efficiency, helping industries identify inefficiencies and root causes of production losses [16, 23]. Automated systems with microcontrollers reduce manual data entry and provide real-time insights into equipment efficiency, allowing operators to quickly address bottlenecks [23].

Intelligent material handling systems help small and medium-sized enterprises improve productivity and sustainability by analyzing equipment efficiency and identifying design limitations [18]. Small-scale food industries face sustainability challenges due to high costs and limited resources. Partnering with experienced suppliers and using combined heat and power systems can help improve affordability and sustainability [12]. Maintenance is essential for drying in paper manufacturing. Improving efficiency and reducing CO_2 emissions is key to sustainability [24]. Improvements in raw materials have further increased sustainability metrics, lowering Eco-indicator 99 values and reducing production costs by 6.2%, while still maintaining product quality [22]. Equipment performance measurement involves downtime, rework, and reliability, requiring skilled operators and adequate supplies. In energy-intensive industries such as brewing, data-driven models support energy management by identifying consumption variations and operational inefficiencies, aiding policy development [19]. Energy analytics further classify non-productive energy consumption into idle and efficiency losses, which are major contributors to excessive energy use in manufacturing systems [29]. Using various analytical tools, manufacturers boost production visibility, uncover hidden capacity, and optimize performance through targeted interventions [32]. This combination of data and lean methodologies provides businesses with actionable insights to achieve sustainable operational improvements [38].

Overall, previous studies demonstrate that lean manufacturing, total productive maintenance, single-minute exchange of die, automation, predictive analytics, and energy management techniques can independently improve production efficiency or reduce environmental impacts. Nevertheless, three important limitations remain. First, OEE improvement and carbon emission assessment have generally been treated as separate research problems. Second, most lean manufacturing studies evaluate productivity-related outcomes without quantifying their environmental implications. Third, empirical evidence linking improvements in equipment availability, performance, and quality to measurable CO_2 emission reductions in beverage production lines is still limited. To overcome these limitations, this research develops an integrated framework that combines OEE measurement, Pareto-based downtime prioritization, fishbone root-cause analysis, and lean manufacturing interventions (total productive maintenance, 5S, and single-minute exchange of die) with electricity-based CO_2 emission assessment. Unlike previous studies that focus either on operational efficiency or on environmental performance, the proposed framework enables simultaneous evaluation of both dimensions using real industrial production data. This integrated approach provides a practical methodology for manufacturers seeking to improve productivity while achieving measurable reductions in carbon emissions and contributes empirical evidence on the operational mechanisms through which lean-driven OEE improvements support industrial decarbonization in the beverage sector.

3 Methods

3.1 Methodological Framework

Figure 1 shows the methodological framework adopted in this study. The flowchart in the Figure illustrates the experimental process from data collection to final results. It visually depicts how data were gathered, analyzed, and interpreted to arrive at the conclusions.

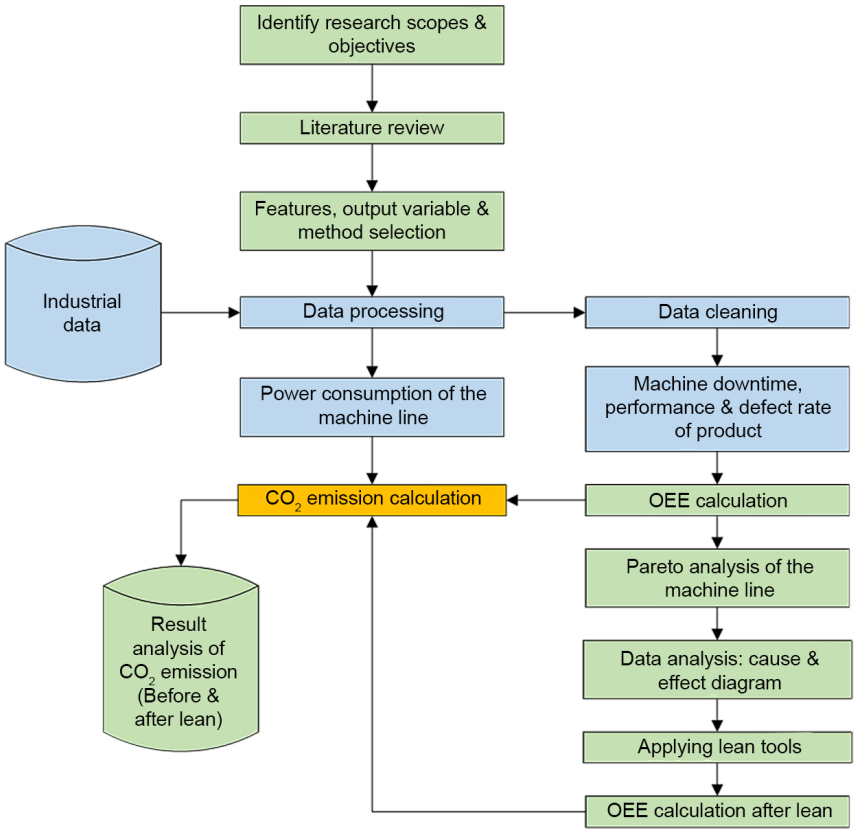


Figure 1. A framework process to improve OEE and reduce CO_2 emissions

Note: OEE = Overall equipment effectiveness; CO_2 = Carbon dioxide.

Table 2. Total power consumption of the machine line

Machine Type	Power Consumption of Machine Line				
	Actual Energy Consumption (kWh)			Average Power Consumption	Share of Total Consumption (%)
	A1	A2	A3		
Premix	1,596.96	1,805.52	1,426.32	267.07	11.28%
Blowmax machine	3,404.16	3,844.80	3,014.64	142.55	23.97%
Blowmax chiller	1,434.24	1,613.52	1,280.40	60.11	10.11%
Filling machine	777.60	822.00	652.56	31.28	5.26%
Capping machine	103.68	117.36	93.36	4.37	0.73%
Flat conveyor & air-conditioning system	933.12	1,061.76	838.80	39.36	6.62%
Air conveyor	984.96	1,107.84	891.12	41.44	6.97%
Warmer-cooler	1,330.56	1,534.56	1,182.24	56.21	9.45%
Labeling machine	466.56	543.12	429.36	19.99	3.36%
Bandroll machine (sleeve)	708.48	790.56	653.76	29.90	5.03%
Date-coding machine	60.00	84.96	66.96	2.94	0.49%
Wrapping machine	2,315.52	2,669.52	2,070.24	97.99	16.48%
Filltech machine	30.96	39.60	30.72	1.41	0.24%
Total	**14,146.80**	**16,035.12**	**12,630.48**	**594.62**	**100.00%**

Machine operators gathered data on downtime to ensure accuracy. The actual available time for the production line was calculated by subtracting downtime from the planned operating time. The machine availability ratio was derived from the time measured between loading and operating durations. To evaluate machine performance, actual production output was assessed and compared to the installed line capacity. Finally, product quality was determined by counting the total number of good final products. OEE was evaluated through the metrics of availability, performance, and quality, which reflect the overall health of the industry. CO_2 emissions were determined using OEE and power consumption data, collected by measuring the total power usage of the production line (Table 2). OEE was higher following the implementation of lean manufacturing tools that prioritize waste reduction, value addition, and continuous improvement. Initially, the challenges affecting the production line were identified using a Pareto analysis, followed by the creation of cause-and-effect diagrams for each specific issue. After the implementation period, CO_2 emissions were recalculated and the data from before and after the lean practices were compared.

3.2 Overall Equipment Effectiveness

Nakajima [39] defined six types of equipment losses. Figure 2 shows the six big losses of OEE. The first two types of losses are downtime losses, which impact a machine's availability ratio. The third and fourth losses are speed losses, which affect the performance ratio when the machine is not operating at its optimal level.

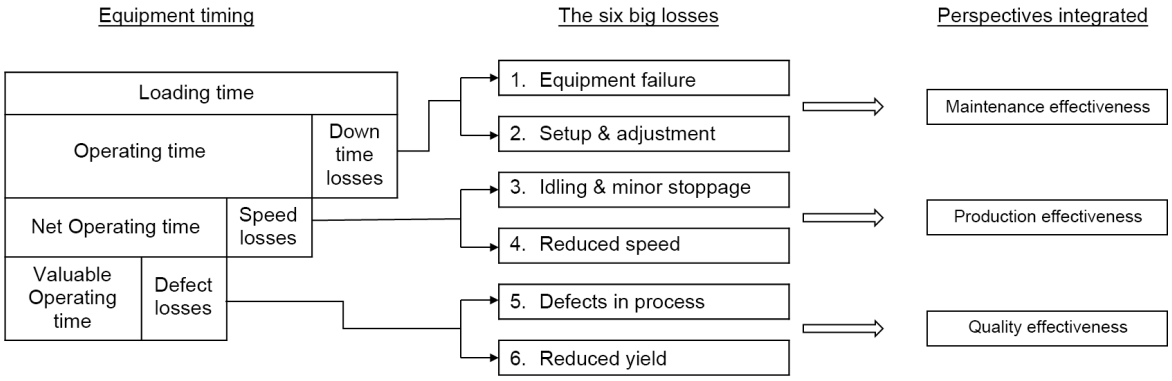


Figure 2. Six big losses of overall equipment effectiveness

The final two losses pertain to defects: an increase in defects results in a lower quality ratio of the produced parts.

3.2.1 Availability

Availability refers to a relationship that indicates the available time for machine operation; this value measures the effectiveness of maintenance activities. The following factors are critical to the availability calculation:

$$\text{Operating time} = \text{Loading time} - \text{Downtime} \tag{1}$$

where, Downtime is the breakdown-induced machine downtime, and Loading time is the amount of time the machine is planned to be in operation.

The availability ratio is calculated as:

$$\text{Availability ratio} = ((\text{ Operating time }) \div (\text{ Loading time })) \times 100\% \tag{2}$$

3.2.2 Performance

Performance measures how efficiently a machine produces products during its operating time. It is evaluated by comparing the actual production output with the ideal output that could have been achieved during the available operating time. The key factors required to calculate performance are:

$$\text{Ideal production rate (products/hour)} = \text{Rated (installed) line capacity} \tag{3}$$

$$\text{Planned production} = \text{Operating time (hour)} \times \text{Ideal production rate (products/hour)} \tag{4}$$

$$\text{Actual production} = \text{Number of products manufactured during the operating time} \tag{5}$$

$$\text{Performance ratio} = ((\text{ Actual production }) \div (\text{ Planned production })) \times 100\% \tag{6}$$

Operating time is recorded in minutes in Table 3 and Table 4 and converted to hours when used in Eq. (4) and Eq. (9).

Table 3. Actual OEE and Carbon dioxide (CO_2) emissions values calculated for different shifts (before lean)

No.	Shift	Loading Time (min)	Operating Time (min)	Availability Ratio	Performance Ratio	Quality Ratio	OEE	CO_2 Emissions per Product (g CO_2/product)
1	Day	840	485	57.74%	52.92%	89%	27.19%	19.35
2	Day	720	591	82.08%	58.45%	89%	42.07%	17.52
3	Night	660	505	76.52%	70.13%	87%	46.69%	14.60
4	Night	675	600	88.89%	72.60%	91%	58.72%	14.11
Total/Average (Shifts 1–29)	—	17,760	13,321	75.01%	67.64%	89%	45.15%	15.14

Note: OEE = overall equipment effectiveness; "—" indicates that the value is not applicable to the aggregated data for multiple shifts.

Table 4. Actual OEE and Carbon dioxide (CO_2) emissions values calculated for different shifts (after lean)

No.	Shift	Loading Time (min)	Operating Time (min)	Availability Ratio	Performance Ratio	Quality Ratio	OEE	CO_2 Emissions per Product (g CO_2/product)
1	Day	720	561	77.92%	75.76%	95%	56.08%	13.52
2	Day	720	524	72.78%	73.47%	94%	50.26%	13.94
3	Night	720	606	84.17%	81.78%	95%	65.39%	12.52
4	Night	540	416	77.04%	71.67%	94%	51.90%	14.29
Total/Average (Shifts 1–29)	—	18,310	14,208	77.60%	75.00%	93%	54.12%	13.65

Note: OEE = overall equipment effectiveness;"—" indicates that the value is not applicable to the aggregated data for multiple shifts.

3.2.3 Quality rate

Quality rate measures the proportion of products that meet the required quality standards. It is calculated by comparing the number of good-quality products with the total number of products produced. The key factors required for calculating the quality rate are process amount (number of products manufactured) and defect (number of defective products).

$$\text{Quality ratio} = (\text{Process amount - Product defect}) \div (\text{Process amount}) \times 100\% \tag{7}$$

3.2.4 Overall equipment effectiveness

After calculating availability, performance, and quality rate, OEE can be determined as follows:

$$\text{OEE} = \text{Availability ratio} \times \text{Performance ratio} \times \text{Quality ratio} \tag{8}$$

OEE is an effective method for assessing the efficiency of a single machine or an integrated machinery system [12]. World-class performance is typically defined by an OEE benchmark of 85%. OEE should exceed 90% for continuous-discrete processes, while for continuous on-stream processes, industries should strive for OEE values of 95% or higher.

3.3 Power Consumption

Power consumption data were gathered from the production line to identify trends in energy usage and to calculate the associated CO_2 emissions. The CO_2 emissions associated with electricity consumption were quantified using a grid emission factor of 620.0 g CO_2/kWh [40]. This indicates that approximately 620.0 g of CO_2 are emitted for each kilowatt-hour (kWh) of electricity consumed. By understanding the CO_2 emissions linked to electricity usage, more informed choices regarding energy consumption can be made.

Power consumption data were collected using digital energy meters installed on each machine and integrated with the plant's Supervisory Control and Data Acquisition (SCADA) system, which continuously logged real-time readings. Measurements were taken in both high- and low-production scenarios to cover the full operational range. To address fluctuations in power readings, three representative values were extracted for each machine: the highest (A2), lowest (A3), and middle (A1) levels. Each of these values was calculated as the average of readings recorded in close proximity to the respective point. This method was employed to minimize the impact of transient fluctuations while still reflecting the observed operating range, ensuring that the highest and lowest recorded values accurately frame the actual power consumption during production. Since the power consumption data were later used to calculate CO_2 emissions, this bounding technique guarantees that the estimated emissions remain within a realistic and representative range, rather than being distorted by temporary spikes or drops in the SCADA-logged data.

As shown in the table, the average power consumption of the production line is 594.62 kW, and the grid emission factor is 620.0 g CO_2/kWh [40].

$$\begin{aligned} CO_2 \text{ emissions per product} = \big(&\text{Operating time per hour} \times \text{Average power consumption} \\ &\times CO_2 \text{ produced per kWh}\big) \div \text{Actual production} \end{aligned} \tag{9}$$

It is important to note that the average power consumption value of 594.62 kW used in Eq. (9) was obtained from three actual load measurements (A1–A3, Table 2) recorded in different shifts, with A2 representing the highest load and A3 the lowest, while the machine line was actively engaged in production, rather than from separate idle or standby readings. Since the idle-state power draw was not isolated from the active-state draw with the available instrumentation, this study assumes that hourly power consumption remains approximately constant throughout the operating time window for the purpose of emissions calculation, as this reflects the average of the observed high, low, and middle load conditions rather than short-term fluctuations within any single hour. This approach serves as a simplifying assumption rather than a measured distinction, highlighting the need for real-time, machine-level metering in future work.

3.4 Schematic Presentation of the Beverage Production Line

Figure 3 shows the industrial beverage production line analyzed in this study, offering an overview of the processing stages evaluated for power consumption and production analysis.



Figure 3. An overview of the beverage bottling production line used in the study

The beverage industry's manufacturing process is fully automated. Machines handle everything from syrup production to packaging. Various other units, including the power plant, compressor, CO_2 plant, and storehouse, support the packaging line. The beverage packaging line begins with a blow molding machine that converts polyethylene terephthalate preforms into bottles using compressed air and heat. The bottles are transported via an air conveyor to the filling machine, where a mixture of syrup, water, and CO_2 is added and sealed. After filling, the bottles go through inspection, warming, labeling, and wrapping before being manually palletized for warehouse storage.

4 Result and Discussion

4.1 Overall Equipment Effectiveness Performance and Downtime Analyses

Data from all 29 shifts, including both day and night operations, were recorded and analyzed in a spreadsheet to determine the OEE and CO_2 emissions per product, both before and after the lean interventions. The averages presented in Tables 3 and 4 were calculated from the complete 29-shift dataset in each case. To illustrate the variability in operational performance, four representative shifts two from the day and two from the night were selected based on the highest and lowest OEE values, rather than exhaustively including all shifts.

Table 3 shows that the average OEE of the production line is 45.15%, with an availability rate of 75.01%, a performance rate of 67.64%, and a quality rate of 89%. Shift 1 (day) had the lowest OEE at 27.19%, while Shift 4

(night) had the highest at 58.72%. The average CO_2 emissions per product were 15.14 g CO_2/product, and lower OEE values were observed alongside higher emissions in the recorded shifts. This OEE is significantly below the world-class benchmark of 85%, highlighting substantial operational losses. A Pareto analysis of machine downtime was performed to identify major contributors to these losses, as shown in Figure 4.

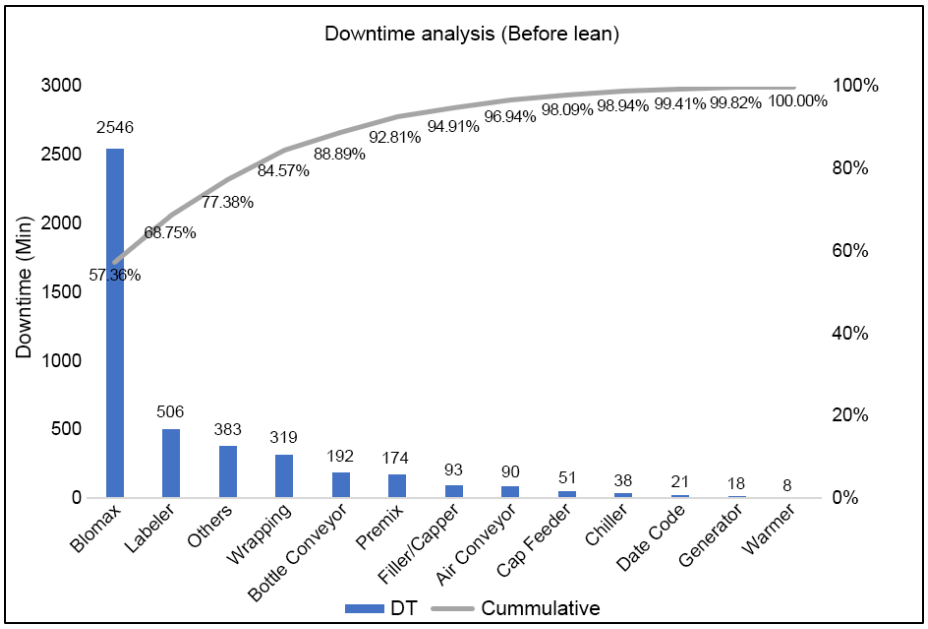


Figure 4. Pareto chart illustrating the distribution of failures along the beverage production line (before lean)
Note: DT = Downtime.

The Pareto chart shows that the Blomax and labeling machines account for nearly 70% of total downtime, highlighting them as primary bottlenecks. Additional downtime in the wrapping and bottle conveyor systems is mainly due to raw material shortages, equipment malfunctions, and operational issues. These findings indicate that excessive machine downtime was associated with the low OEE of the production line. Lean manufacturing tools such as total productive maintenance and 5S were implemented to address equipment reliability and operational practices; subsequent changes in production efficiency and CO_2 emissions were then evaluated.

4.2 Cause-and-Effect Diagram of Overall Equipment Effectiveness Losses

The cause-and-effect diagram in Figure 5 shows several issues impacting OEE in the beverage processing line and determines the root causes of machine downtime. Significant concerns during the blowing stage were critical to the beverage production line, and malfunctions led to significant downtime. Issues such as faulty heating, inadequate cooling, and the production of defective bottles were common. Additionally, non-functioning sensors, improper mold closure, and inadequate maintenance of outdated parts contributed to downtime challenges. Material supply was critical for production, but quality issues with preforms caused machine stoppages in blow molding. Recycled preforms adversely affected bottle quality, and air shortages led to explosions. Noise and vibration from machines created health risks for employees. The management system needed improvements in maintenance, training, and safety oversight. The filling machine experienced challenges such as underfilling and overfilling due to issues with the solenoid valve and flow meter, which resulted in syrup loss. Inadequate bottle sealing stemmed from a worn gripper, a faulty neck spray, and problems with the capper closure. Proper adjustment of the solenoid valve was crucial to avoid these filling discrepancies. Additionally, the use of poor-quality bottles led to waste, while inefficient shifts negatively affected employee productivity.

The conveyor system comprised a blowing stage and an air conveyor. Malfunctions in the air conveyor guide caused bottle instability and subsequent falls from the conveyor, while sensor issues on the belt conveyor hindered speed adjustments. Frequent motor malfunctions caused jamming, and improper lubrication led to premature wear of the bottles. Additionally, aging components further exacerbated these challenges. The packaging stage consisted of labeling, date coding, wrapping, and palletizing. Alarms were triggered by high temperatures in the wrapping machine due to electronic issues, which negatively impacted product quality. Maintaining the correct moisture level was essential for the new labels to prevent tearing. Unfortunately, inadequate glue and low-quality materials led to misaligned date codes and tearing, while high-quality shrink film effectively supported the weight of the cases. The manual palletizing system increased labor requirements, often causing damage to entire cases at the end of wrapping because they were not placed on the trolley in time, which created bottlenecks in the entire line.

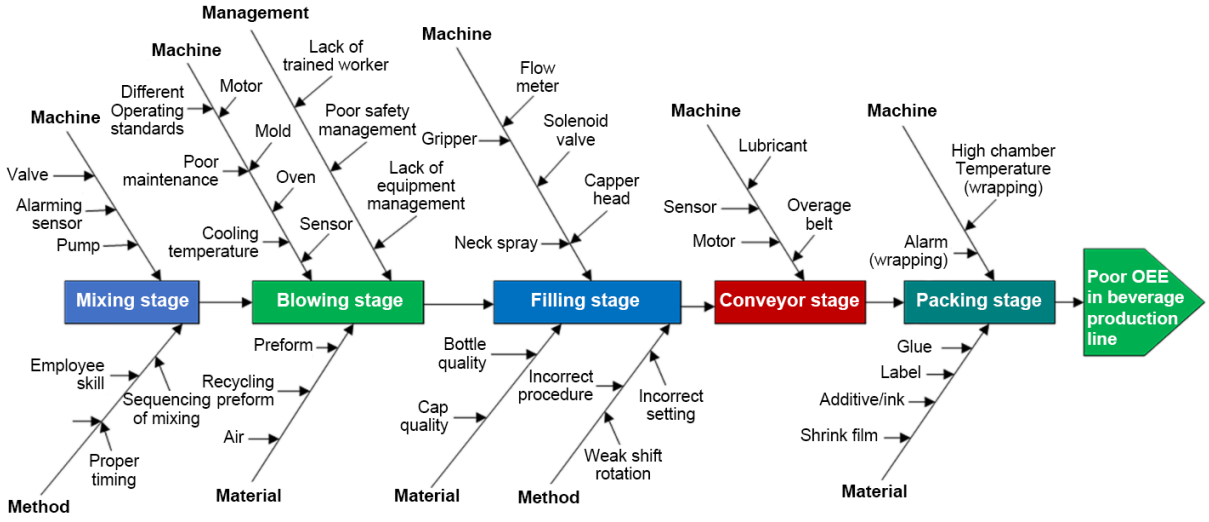


Figure 5. Fishbone diagram for poor OEE in the beverage production line

Note: SOP = Standard Operating Procedure; OEE = Overall Equipment Effectiveness.

4.3 Lean Improvement Strategies

All issues with identified root causes are difficult to resolve, but using lean tools can significantly reduce many problems and enhance operational efficiency. The layout of the beverage production line and the major identified problems and solutions are shown in Figure 6. The initial analysis revealed factors contributing to low OEE, including downtime of the Blomax blow molding machine, congestion in the filling and capping sections, electronic failures in the wrapping machine, and limited capacity in the manual palletizing area. The filler dashboard also indicated a bottleneck due to inconsistent bottle flow.

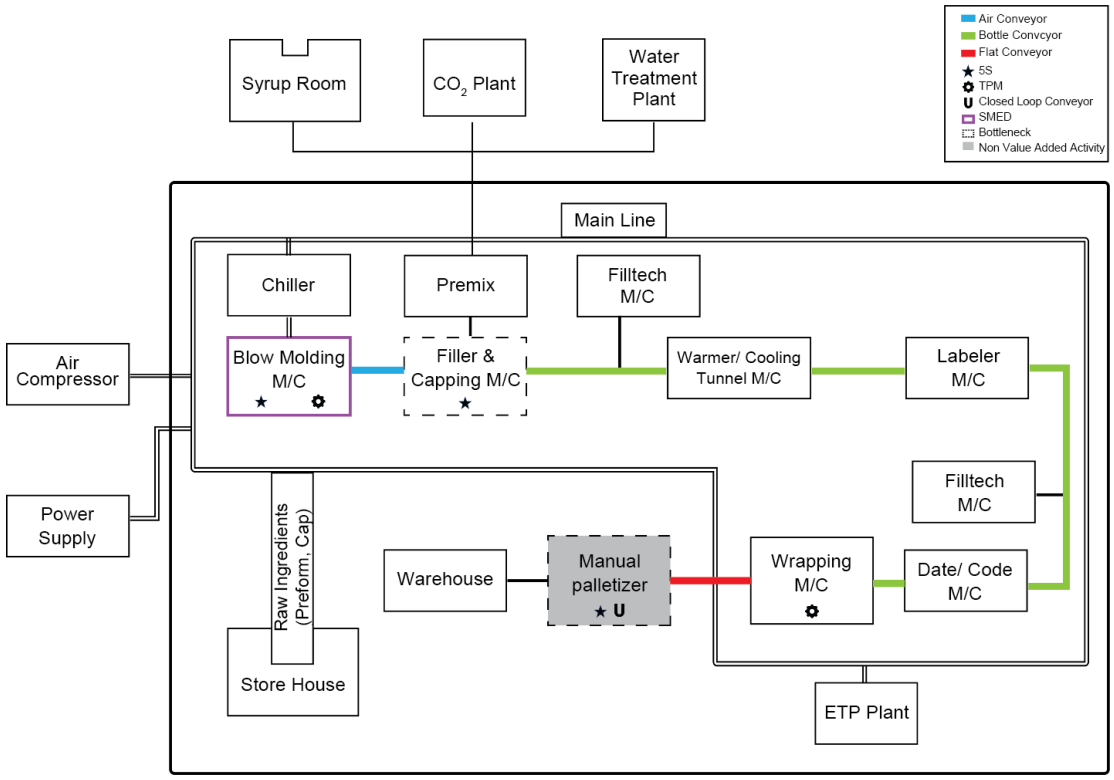


Figure 6. Beverage production line layout focusing on bottlenecks and lean-based improvement strategies for overall equipment effectiveness enhancement

Note: 5S = Sort, Set in Order, Shine, Standardize, and Sustain; TPM = Total Productive Maintenance; SMED = Single-Minute Exchange of Die; CO_2 = Carbon Dioxide; M/C = Machine; ETP Plant = Effluent Treatment Plant.

To address these issues, several lean manufacturing tools were implemented. The single-minute exchange of die technique was optimized for mold preparation and changeover activities in the blow molding process, as shown in Figure 7.

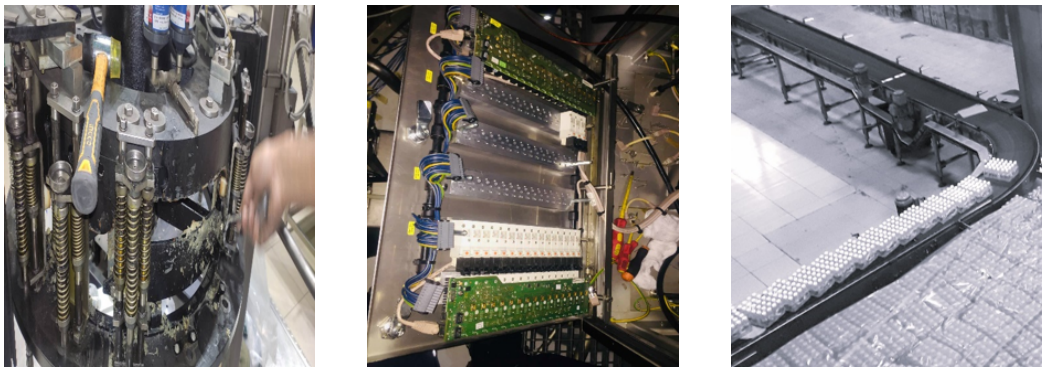


(a) Installed mold unit (b) Mold components (c) Assembled mold cavity

Figure 7. Preparation and assembly of blow-molding molds during single-minute exchange of die-based rapid changeover



(a)



(b) (c) (d)

Figure 8. Lean manufacturing improvement activities involving Sort, Set in Order, Shine, Standardize, and Sustain (5S), total productive maintenance, and bottleneck reduction: (a) total productive maintenance-based autonomous maintenance of Blomax parts; (b) operator lubricating the filler as part of 5S (Shine) (c) troubleshooting of the wrapping machine (d) u-loop conveyor.

Total productive maintenance was applied to the Blomax machine and wrapping system to minimize failures and enhance availability, as illustrated in the lean manufacturing improvement in Figure 8. Additionally, the 5S organization was implemented in the filler section for better process visibility. Organizing tools, spare parts, and measurement devices ensured that solenoid valves, flow meters, cappers, and grippers were easily

accessible and properly maintained. To address a bottleneck in manual palletizing, a U-loop conveyor system was introduced to reduce non-value-added time, enabling multiple trolleys to transport finished cases simultaneously. These improvements enhanced process flow, reduced downtime, and increased overall production efficiency and effectiveness.

Additionally, all operators received comprehensive training to operate machinery safely and efficiently. This training covered essential operations, safety protocols, machine repair, and basic engine maintenance. Participants were given training cards with troubleshooting procedures and rewards to help them resolve technical issues independently. The 240-day implementation period provided key opportunities for advancement. Initially, some operators were hesitant to adopt the standardized 5S and total productive maintenance routines, viewing them as extra workload. On-the-floor coaching facilitated a smoother transition to compliance, while training improved operators' skills despite a temporary reduction in available labor. Important modifications were also made, including installing a U-loop conveyor to enhance bottle transfer and reduce jams at the labeler, which required upfront investment and downtime but was essential for optimizing operations. By the end of the period, consistent improvements in OEE and emissions were observed, highlighting the gradual nature of lean-driven decarbonization supported by effective change management.

4.4 Impact of Lean Implementation on Overall Equipment Effectiveness and Carbon Dioxide Emissions

Table 4 displays the OEE performance of the beverage production line following the implementation of lean improvement strategies. As with Table 3, the four shifts shown represent the maximum and minimum OEE shifts (two day, two night) selected from the complete 29-shift post-lean dataset (see note above Table 3).

Data were collected after 240 days of applying these lean practices, focusing on machine downtime and availability. The results indicate a significant improvement in OEE, which increased from 45.15% to 54.12%. The increase in OEE coincided with a decrease in CO_2 emissions. However, additional implementation time and ongoing improvement efforts are necessary to attain even higher OEE performance. The analysis of data revealed a significant decrease in CO_2 emissions alongside the increase in OEE following the implementation of lean practices. Figure 9 graphically represents the relationship between CO_2 emissions and the three key factors: performance rate, availability rate, and quality rate. Figure 9a shows an inverse relationship between the performance ratio and CO_2 emissions (g CO_2/product), with higher observed performance ratios coinciding with lower emissions. During one shift, CO_2 emissions peaked at 19.35 g CO_2/product while the performance ratio dropped to 52.92%.

Figure 9b illustrates an inverse relationship between CO_2 emissions and machine availability: as availability increases, emissions generally decrease. Therefore, the observed association suggests that periods with less machine downtime coincided with lower emissions. The relationship between production rate and machine availability provides valuable insights into the factors associated with CO_2 emissions. The production rate demonstrates a clear inverse relationship with CO_2 emissions, with emissions decreasing from 19.35 to 12.15 g CO_2/product as the production rate increased from 52.92% to 84.31%. This pattern is consistent with lower emissions per product when more products were produced during the observed operating conditions, although the before–after design does not establish a direct causal effect. Conversely, the availability rate demonstrates greater variability. For instance, after lean implementation, an availability rate of 78.64% was associated with emissions of 12.50 g CO_2/product, whereas an availability rate of 84.17% was associated with 12.52 g CO_2/product. This relatively small variation suggests that availability alone does not fully determine emissions, as different downtime events can affect operating time and idle power consumption differently. Therefore, although minimizing downtime remains an operational objective, the observed data indicate that performance losses, such as reduced operating speed and minor stoppages, were also associated with variation in CO_2 emissions per product.

Changes following lean manufacturing implementation included higher observed machine performance and overall production efficiency alongside lower CO_2 emissions. Specifically, average emissions per product decreased from 15.14 to 13.65 g CO_2/product. Figure 9c further shows an inverse relationship between product quality and CO_2 emissions, where higher product quality was associated with lower emissions. Optimized machines and processes were observed alongside higher-quality production, whereas lower quality may increase material waste, rework, and the resources required to meet production demand. The implementation of lean manufacturing tools addressed issues such as overfilling and product defects, with product quality increasing from 89% to 93% alongside the observed reduction in CO_2 emissions. Figure 9d presents the relationship between OEE and CO_2 emissions before and after the implementation of lean manufacturing tools. Following implementation, the maximum CO_2 emissions per product decreased from 19.35 to 14.31 g CO_2/product, and the minimum was about 12.50 g CO_2/product. These findings indicate that lean manufacturing tools were associated with simultaneous changes in OEE and CO_2 emissions, highlighting their potential to support both production efficiency and environmental performance.

Figure 10 compares CO_2 emissions across production shifts before and after the implementation of lean manufacturing tools. Before using lean tools, CO_2 emissions exceeded 16.00 g CO_2/product in nearly every shift. However, after implementing lean tools, the CO_2 emissions for each shift ranged from 12.50 to 14.35 g

CO_2/product, coinciding with the observed improvement in OEE.

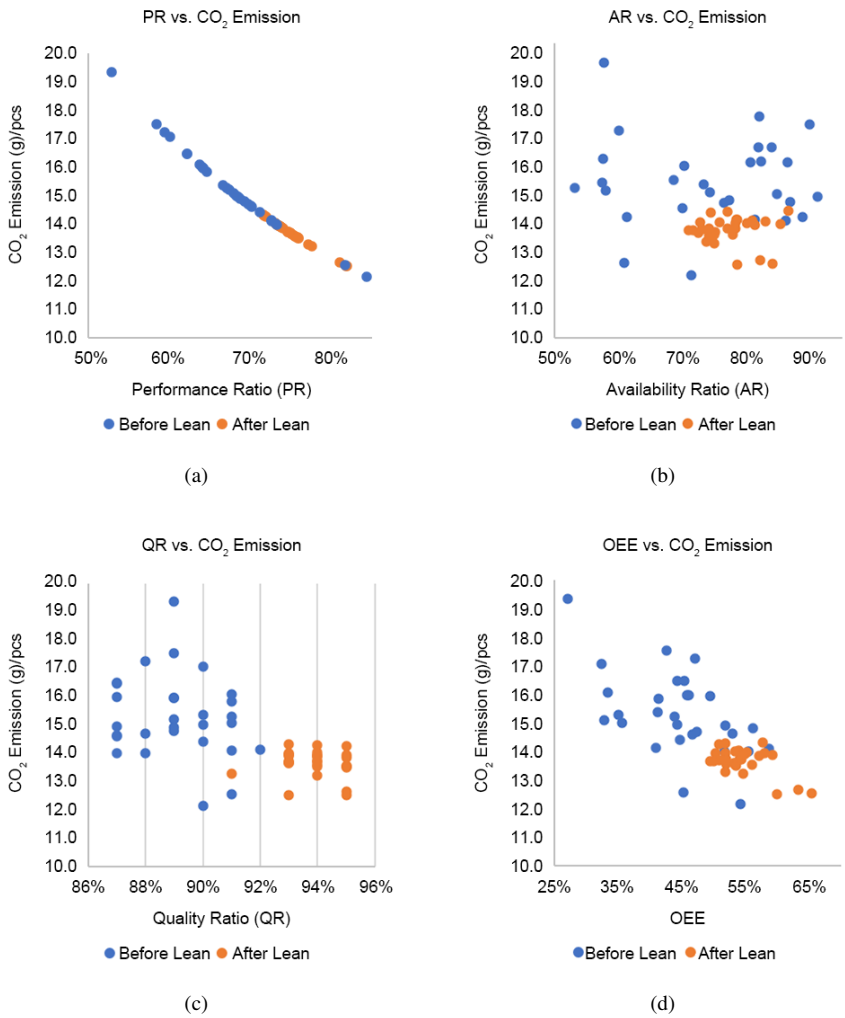


Figure 9. Comparison of CO_2 emissions with OEE and its components before and after lean implementation for the beverage production line: (a) performance ratio versus CO_2 emissions; (b) availability ratio versus CO_2 emissions; (c) quality ratio versus CO_2 emissions; (d) OEE components versus CO_2 emissions

Note: Emission (g/pcs) = Carbon Dioxide Emissions per Piece of Product; OEE = overall equipment effectiveness; CO_2 = Carbon dioxide.

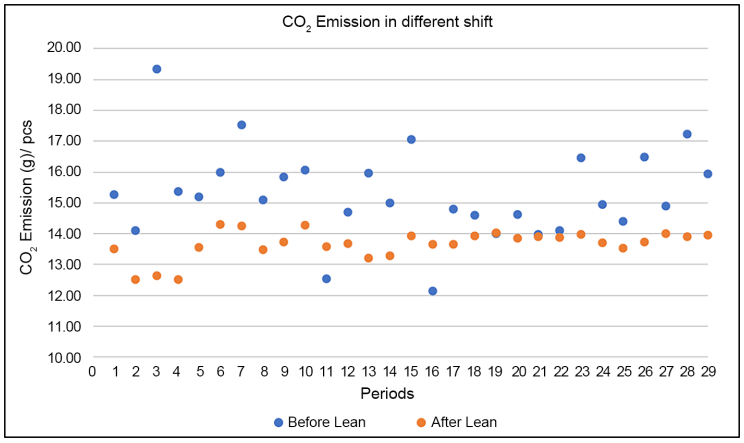


Figure 10. Shift variation of CO_2 emissions rate before and after lean

Note: CO_2 = Carbon dioxide.

Figure 11 shows CO_2 emissions for day and night shifts in the beverage production line. The night shift had the highest OEE at 65.39% and the lowest CO_2 emissions at 12.52 g CO_2/product. In contrast, the day shift had comparatively lower OEE and higher emissions. This difference coincided with fewer machine failures during the night shift and higher observed operational efficiency; the data do not establish whether the lower failure frequency directly caused the lower emissions.

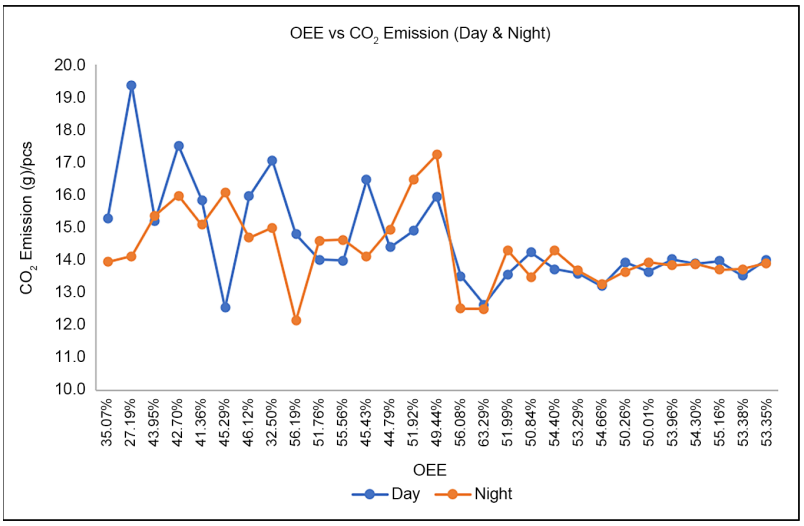


Figure 11. Comparison of OEE and CO_2 emissions between day and night shifts

Note: OEE = overall equipment effectiveness; CO_2 = Carbon dioxide.

5 Conclusion

This study examined the relationship between OEE and CO_2 emissions in a beverage manufacturing line before and after the implementation of lean manufacturing practices. Following the implementation of total productive maintenance, 5S, and standardized training protocols, OEE increased from 45.15% to 54.12%, while per-unit CO_2 emissions decreased by 9.8%, from 15.14 to 13.65 g CO_2/product. Improvements were particularly observed in machine performance and availability, accompanied by reductions in downtime in critical units such as the Blomax and labeling machines. These observed changes indicate an association between improved equipment effectiveness, reduced operational losses, and lower CO_2 emissions per product. However, given the before–after case study design without a control group, these findings should not be interpreted as evidence of a direct causal relationship. The observed improvements were achieved primarily through low-cost procedural lean practices rather than substantial capital investments. The implementation of 5S and total productive maintenance involved labor reallocation, employee training, and structured maintenance scheduling, with only minor physical modifications, including single-minute exchange of die-type changeovers and a U-loop conveyor adjustment. During the study period, these practices coincided with an increase of 8.97 percentage points in OEE and a reduction of 1.49 g CO_2/product in average emissions. This relatively low capital requirement suggests that lean practices may provide a practical approach for manufacturers in resource-constrained environments such as Bangladesh. Nevertheless, other operational or contextual factors may have contributed to the observed changes. This study has several limitations. Data were collected from a single machine line within one manufacturing facility, which limits the generalizability of the findings. Variations in OEE across shifts may also reflect differences in workers' experience and problem-solving skills. In addition, CO_2 emissions associated with electricity consumption were estimated using an average grid emission factor, which may vary according to the regional energy mix and consequently affect emission estimates. Most importantly, the absence of a control group and the before–after study design limit the ability to isolate the effects of lean practices from other factors occurring during the study period.

Future research could employ IoT-enabled sensor arrays to monitor machine-level energy consumption, downtime, and operating conditions in real time. Integrating these data with manufacturing execution systems and enterprise resource planning platforms could support more detailed OEE–CO_2 monitoring and predictive maintenance. Studies involving multiple production lines, facilities, and manufacturers, together with control or comparison groups and longer observation periods, would provide stronger evidence regarding the relationship between lean practices, OEE, and CO_2 emissions. Overall, the findings suggest that improvements in OEE were accompanied by increased production efficiency and reduced CO_2 emissions per product, highlighting the potential association between lean manufacturing practices and concurrent operational and environmental performance improvements.

Author Contributions

Conceptualization, T.A.K.; Methodology, T.A.K.; Software, M.Z.H.C.; Formal Analysis, T.A.K.; Investigation, T.A.K.; Data Curation, T.A.K.; Writing—Original Draft Preparation, T.A.K. and M.Z.H.C.; Writing—Review and Editing, N.A.S.; Visualization, M.Z.H.C.; Supervision, S.M.A.; Project Administration, S.M.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this work, the authors used QuillBot Premium/Grammarly Premium to improve the quality of the writing and check for grammatical errors. After using these tools/services, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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